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HBM25070

Business Statistics Assignment

1)

CLASS INTERVAL	Frequency (F)
0 – 10	10
10 – 20	x
20 – 30	25
30 – 40	y
40 – 50	10
Total	80

Total frequency = 80
Adding all

$10 + x + 25 + y + 10 = 80$

$45 + x + y = 80$

$x + y = 35$

$$\text{Mean} = \frac{\Sigma(f \cdot x)}{\Sigma f}$$

Midpoint = (lower limit + upper limit) ÷ 2

CLASS INTERVAL	Frequency (F)	Midpoint (M)
0 – 10	10	5
10 – 20	x	15
20 – 30	25	25
30 – 40	y	35
40 – 50	10	45

$\Sigma fm = 10 \cdot 5 + x \cdot 15 + 25 \cdot 25 + y \cdot 35 + 10 \cdot 45$

$\Sigma fm = 50 + 15x + 625 + 35y + 450$

$\Sigma fm = 1125 + 15x + 35y$

$$26 = \frac{1125 + 15x + 35y}{80}$$

$$26 \times 80 = 1125 + 15x + 35y$$

$$2080 = 1125 + 15x + 35y$$

$$955 = 15x + 35y$$

$$191 = 3x + 7y$$

$x + y = 35$

$3x + 7y = 191$

$$x = 35 - y$$

$$3(35 - y) + 7y = 191$$

$$105 - 3y + 7y = 191$$

$$105 + 4y = 191$$

$$4y = 86$$

$$y = 21.5$$

$$x = 35 - y = 35 - 21.5 = 13.5$$

$$x = 13.5$$

$$y = 21.5$$

2)

CLASS INTERVAL	Frequency (F)
0-20	5
20-40	x
40-60	30
60-80	y
80-100	10
Total	70

Median = 46

$5 + x + 30 + y + 10 = 70$

$x + y = 25$

$$\text{Median} = l + \left(\frac{\frac{N}{2} - CF\,Prev}{F}\right) h$$

Median $= 46$

$l = 40$

$N/2 = 35$

$cf(\text{Previous}) = 5 + x$

$f = 30$

$h = 20$

$$46 = 40 + \left(\frac{35 - (5 + x)}{30}\right) 20$$

$46 - 40 = 6$

$$6 = \frac{30 - x}{30} \times 20$$

$$6 = \frac{2}{3}(30 - x)$$

$18 = 2(30 - x)$

$9 = 30 - x$

$x = 21$

$x + y = 25$

$21 + y = 25$

$y = 4$

$x = 21$

$y = 4$

- Calculate Q_3 (Third Quartile)

$$Q_3 = \frac{3}{4}(70) = 52.5 \text{ Value}$$

$L = 40$

cf Prev $= 5 + 21 = 26$

$f = 30$

$h = 20$

$$Q_3 = 40 + \left(\frac{52.5 - 26}{30}\right) 20$$

$$= 40 + \left(\frac{26.5}{30}\right) 20$$

$$40 + 17.67$$

$$Q_3 = 57.67$$

P_{77} (77^{th} Percentile)

$$P_{77} = 0.77 \times 70 = 53.9$$

$$P_{77} = 40 + \left(\frac{53.9 - 26}{30}\right) 20$$

$$= 40 + \left(\frac{27.9}{30}\right) 20$$

$$= 40 + 18.6$$

$$P_{77} = 58.60$$

$x = 21$

$y = 4$

$Q_3 = 57.67$

$P_{77} = 58.60$

3)

Wages	F	Midpoint(m)	Fm
200-300	5	250	1250
300-400	8	350	2800
400-500	12	450	5400
500-600	15	550	8250
600-700	10	650	6500

$1250 + 2800 + 5400 + 8250 + 6500 = 24200$

$$\bar{x} = \frac{24200}{50} = 484$$

m	F	d=m-484	d^2	$F \cdot d^2$
250	5	-234	54756	273780
350	8	-134	17956	143648
450	12	-34	1156	13872
550	15	66	4356	65340
650	10	166	27556	275560

$273780 + 143648 + 13872 + 65340 + 275560 = 772,200$

$$\sigma = \sqrt{\frac{\Sigma f d^2}{N}}$$

$$\sigma = \sqrt{\frac{772200}{50}}$$

$$= \sqrt{15444}$$

$$= 124.2 \text{ (Approx)}$$

4) Coefficient of variation

$$CV = \frac{\sigma}{\bar{x}} \times 100$$

CV Plant A

$$CV_A = \frac{25}{250} \times 100$$

$$CV_A = 0.1 \times 100 = 10\%$$

CV Plant B

$$CV_B = \frac{30}{280} \times 100$$

$$CV_B = 0.1071 \times 100 = 10.71\%$$

Plant A = 10%
Plant B = 10.71%

5) Correlation Calculation
$r = 0.997$ (Approx)

- The correlation between aptitude test scores and job performances rating is approx 0.997 very close to 1.

- This means their is extremely strong positive relationshi between two.

- The Aptitude test is highly useful for predicting first year job performance.

6) Spearman's rank correlation

$$\rho = 1 - \frac{6 \sum d^2}{n(n^2-1)}$$

Feature	Group X (R_1)	Group Y (R_2)	$d = R_1 - R_2$	d^2
A	3	3	0	0
B	5	4	1	1
C	6	5	1	1
D	1	1	0	0
E	4	7	-3	9
F	7	5	2	4
G	1	2	-1	1

$$\sum d^2 = 0+1+1+0+9+4+1 = 16$$

$$\rho = 1 - \frac{6 \times 16}{7(7^2-1)}$$

$$= 1 - \frac{96}{7 \times 48}$$

$$= 1 - \frac{96}{336}$$

$$1 - 0.2857$$

$$\rho = 0.7143$$

$$F = \frac{s^2 A}{s^2 B}$$

$$F = \frac{9.8}{4.2} = 2.3333$$

$$F = 2.33$$

Critical F value

- $df_1 = n_A - 1 = 21 - 1 = 20$
- $df_2 = n_B - 1 = 16 - 1 = 15$

Level of significance $a = 0.05$

Right tail test $= (0.05, 20, 15)$

$$F \approx 2.28$$

Calculated

$F = 2.33$

Critical

$F = 2.28$

Since 2.33 is bigger value then 2.28 it mean

yes portifolio A really has a significantly more ris than portfolio B at 5% level.

Portfolio A is riskier than portfolio B.

9) $H_0 : \mu = 10.0$
$H_1 : \mu \neq 10.0$
two tailed test

- Test statistic

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$

$\bar{x} = 9.85$
$\mu = 10.0$
$s = 0.5$
$n = 15$
$\sqrt{n} = \sqrt{15} = 3.872983346$

$$\frac{s}{\sqrt{n}} = \frac{0.5}{3.87298} = 0.129099 \text{ (Approx)}$$

$$t = \frac{9.85 - 10.0}{0.129099} = \frac{-0.015}{0.129099} \approx -1.1625$$

- Critical Value and decision

$t = 2.977$ (i.e $t = 0.005, 14 \approx 2.977$)

$t = 1.1625$
$t = 1.1625 < 2.977$

$P = 0.264$

- Confidential interval

99% CI for the mean $\bar{x} \pm t_{0.005,14} \frac{8}{\sqrt{n}}$

$9.85 \pm 2.977 \times 0.129099 \approx 9.85 \pm 0.384$

So $CI = (9.466, 10.234)$ which includes 10

= So we do not have enough evidence to say the ~~the~~ true mean lenght is different from 10 cm

10)

$$Z = \frac{\bar{x} - \mu_0}{\sigma / \sqrt{n}}$$

$$\sigma/\sqrt{n} = \frac{5}{\sqrt{36}} = \frac{5}{6} = 0.83333$$

$$Z = \frac{148.5 - 150}{0.833} = \frac{-1.5}{0.833} = 1.80$$

$$Z = 1.80$$

- Critical Value two tailed $(\alpha = 0.01)$

$$Z_{crit} = 2.57581 \text{ (Approx)}$$

Decision Value

$Z < -2.5758$ or $Z > + \cancel{2.58}\ 2.5758 \rightarrow$ reject H_0

$Z = 1.80 < 2.5758$

3) P value

$P = 2 \times P(z < 1.80)$

$P(z < -1.80) = 0.0359$

$P = 2 \times 0.0359 = 0.0718$ (0.072)

4) 99% Confidence Interval

$\bar{x} \pm z_{0.0005} \frac{\sigma}{\sqrt{n}} = 148.5 \pm 2.5758 \times 0.83333$

Margin $= 2.576 \times 0.83333 = 2.15$

$CI = [148.5 - 2.15, 148.5 + 2.15] = [146.35, 150.65]$

CI contain $= 150$

11)

	In Store	Online	Total
Male	40	60	100
Female	30	70	100
Column Total	70	130	200

$E.F = \frac{(\text{Row total})(\text{Column total})}{\text{Grand total}}$

For males

Male in store $\frac{100 \times 70}{200} = 35$

Male Online $\frac{100 \times 130}{200} = 65$

For Females

In Store $= \frac{100 \times 70}{200} = 35$

In Online: $\frac{100 \times 130}{200} = 65$

$$x^2 = \sum \frac{(O-E)^2}{E}$$

- Male in Store

$$\frac{(40-35)^2}{35} = \frac{25}{35} = 0.714$$

$$= 0.714$$

- Male Online:

$$\frac{(60-65)^2}{65} = \frac{25}{65} = 0.385$$

$$0.385$$

- Female in Store:

$$\frac{(30-35)^2}{35} = 0.714$$

$$= 0.714$$

- Female Online

$$\frac{(70-65)^2}{65} = 0.385$$

$$x^2 = 0.714 + 0.385 + 0.714 + 0.385$$

$$= 2.198$$

$$df = (r-1)(c-1) = (2-1)(2-1) = 1$$

$$\chi^2 = 3.84$$

$$\chi^2 = 2.198 < 3.84$$

So we failed to reject the null hypothesis.

12)

Day	Observed (O)
Monday	110
Tuesday	95
Wednesday	80
Thursday	105
Friday	110
Total	500

$$\chi^2 = \sum \frac{(O-E)^2}{E}$$

Each day $(E = 100)$

- Monday
  $(110-100)^2 / 100 = 100 / 100 = 1.00$
- Tuesday
  $(95-100)^2 / 100 = 25 / 100 = 0.25$
- Wednesday
  $(80-100)^2 / 100 = 400 / 100 = 4.00$
- Thursday
  $(105-100)^2 / 100 = 25 / 100 = 0.25$
- Friday
  $(110-100)^2 / 100 = 100 / 100 = 1.00$

Sum:

$$\chi^2 = 1 + 0.25 + 4 + 0.25 + 1 = 6.50$$

Degrees of freedom

$$k - 1 = 5 - 1 = 4$$

Common 5% significance level the critical chi square df = 4 about 9.49

Calculated $\chi^2 = 6.50$
Critical $\chi^2_{0.005, 4} = 9.49$

Since $6.50 < 9.49$ we do not reject hypothesis.

13)

$y = a + bx$

Student	x (hour Studied)	y (exam Score)
1	2	55
2	4	65
3	5	70
4	7	85
5	9	95

Calculating regression line

$\bar{x} = 5.4$
$\bar{y} = 74$

$$b = \frac{\Sigma(x-\bar{x})(y-\bar{y})}{\Sigma(x-\bar{x})^2} = \frac{172}{29.2} = 5.89$$

$a = \bar{y} - b\bar{x} = 74 - 5.89(5.4) = 42.19$

$Y = 42.19 + 5.89X$

Predicting score when $x = 6$ hours

$Y = 42.19 + 5.89(6)$

$Y = 42.19 + 35.34 = 77.53$

The exam score is $= 78$ Marks.

14)

Plot	Standard	New	Difference d = New - Standard
1	58	61	3
2	65	64	-1
3	70	73	3
4	62	68	6
5	55	60	5
6	75	74	-1
7	60	65	5
8	68	69	1
9	59	62	3
10	68	71	3

$N = 10$

$D = 2.7$

$S_d = 2.406$

Mean difference $SE = S_d/\sqrt{n} = 0.761$

$t = \frac{\bar{d}}{SE} = \frac{2.7}{0.761} = 3.55$

$df = n - 1 = 9$ $\quad t_{0.05,9} = 1.833$

3.55 larger than 1.833

$\bar{d} \neq t_{0.025,9}$ $SE = 2.7 \neq 1.72$

$CI = (0.98, 4.42)$ / 1 and 4.4 bushels/acre